

2016 年真题参考答案

一、选择题

(1) B. (2) D. (3) B. (4) B. (5) A. (6) D. (7) C. (8) C.

二、填空题

(9) $y = x + \frac{\pi}{2}$. (10) $\sin 1 - \cos 1$. (11) $y' - y = 2x - x^2$.

(12) $5 \cdot 2^{n-1}$. (13) $2\sqrt{2}v_0$. (14) 2.

三、解答题

(15) $e^{\frac{1}{3}}$.

(16) $f'(x) = \begin{cases} 4x^2 - 2x, & 0 < x < 1, \\ 2x, & x \geq 1. \end{cases}$ 最小值 $f\left(\frac{1}{2}\right) = \frac{1}{4}$.

(17) 点 $(-1, -1)$ 为极大值点, 极大值为 1.

(18) $1 - \frac{\pi}{2}$.

(19) $y = k_1 e^x - k_2(2x + 1)$, 其中 k_1, k_2 为任意常数.

(20) 体积 $\frac{18}{35}\pi$, 表面积 $\frac{16}{5}\pi$.

(21) (I) 平均值为 $\frac{1}{3\pi}$.

(II) 证明略.

(22) (I) $a = 0$.

(II) $k(0, 1, -1)^T + (1, -2, 0)^T$, 其中 k 为任意常数.

(23) (I) $A^{99} = \begin{pmatrix} 2^{99} - 2 & 1 - 2^{99} & 2 - 2^{98} \\ 2^{100} - 2 & 1 - 2^{100} & 2 - 2^{99} \\ 0 & 0 & 0 \end{pmatrix}$.

(II) $\beta_1 = (2^{99} - 2)\alpha_1 + (2^{100} - 2)\alpha_2$, $\beta_2 = (1 - 2^{99})\alpha_1 + (1 - 2^{100})\alpha_2$, $\beta_3 = (2 - 2^{98})\alpha_1 + (2 - 2^{99})\alpha_2$.